Elias Hossain

AI/ML Researcher | Reinforcement Learning, LLM Post-Training, Reliable Agentic Systems

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Research Profile

Doctoral researcher in **reinforcement learning, LLM post-training, and reliable agentic AI**. My work combines theoretical analysis (closed-form results), large-scale experimentation, and uncertainty estimation, and I ship released code and benchmarks alongside each paper. Publications include **NeurIPS 2026 (main track)**, **EACL 2027**, and **Findings of AACL-IJCNLP 2026**, with further work under review.

Education

University of Central Florida, Orlando, FL — Aug 2025 – May 2028 (Expected)
Ph.D. in Industrial Engineering (CS concentration), GPA 3.87/4.00

Mississippi State University, Mississippi State, MS — Jan 2024 – Jul 2025
M.S. in Computer Science (Research), GPA 3.88/4.00

Research and Technical Experience

Graduate Research Assistant — Aug 2025 – Present
Complex Adaptive Systems Laboratory, University of Central Florida — *Orlando, FL*

- Lead research on **post-training policy steering** for offline RL, **inference-time uncertainty calibration**, **runtime safety for tool-using LLM agents**, and **multi-agent security**; six papers submitted or accepted (see Publications).
- Built **multi-agent preference-data verification**, reducing Qwen2-7B summarization refusals from **14.8% to 0.5%** without reducing win rate.
- Released **5 public ML benchmarks** (**74,590+ records**) with code; ran multi-GPU experiments on Qwen, Llama, and Mistral models.

Graduate Research Assistant — May 2025 – Jul 2025
Mississippi State University — *Mississippi State, MS*

- Developed contrastive-learning and uncertainty-estimation methods for detecting rare malicious events in imbalanced security logs.
- Implemented **FAISS** retrieval and **RAGAS** evaluation for BarkPlug V.2, a retrieval-augmented assistant deployed to the campus community.

Software Engineer — Dec 2022 – Nov 2023
REVE Systems — *Dhaka, Bangladesh*

- Reduced speech-recognition **word error rate by 20%** and **inference time by 15%** using DeepSpeech and Wav2Vec 2.0; deployed models in a production e-governance voice interface.

AI/ML Researcher — Oct 2020 – Nov 2022
Time Research & Innovation — *United Kingdom*

- Developed X-ray and CT deep-learning models for COVID-19 screening and ML components for a telemedicine platform.

Selected Publications and Preprints

[1] **When Policies Cannot Be Retrained: A Unified Closed-Form View of Post-Training Steering in Offline Reinforcement Learning**
E. Hossain, *NeurIPS 2026 (main track), accepted* — arXiv:2604.22873
Studies deployment-time adaptation of frozen offline-RL actors via Product-of-Experts composition with a goal-conditioned prior, and proves that PoE and KL-regularized adaptation yield the same policy. Across 4 D4RL environments (3,900 MuJoCo episodes), composition degrades gracefully under poor priors while baselines collapse, and a KL-budget selector often recovers a near-oracle operating point.

[2] **UAT-LITE: Inference-Time Uncertainty-Aware Attention for Pretrained Transformers**
E. Hossain, *EACL 2027, committed* arXiv:2602.02952

Injects epistemic uncertainty into self-attention via Monte Carlo dropout, with no change to pretrained weights or training objective, plus a layer-wise variance decomposition. Reduces expected calibration error by about 20% relative to fine-tuned BERT-base on SQuAD 2.0, MNLI, and SST-2 while preserving accuracy.

[3] **NEXUS: Structured Runtime Safety for Tool-Using LLM Agents**
E. Hossain, *Findings of AACL-IJCNLP 2026, accepted* arXiv:2607.19356

A runtime safety monitor that chooses among allow, block, confirm, or revise using deterministic rules, argument-level inspection, and a calibrated risk score. Reaches 0.949 F1 on a synthetic benchmark, 0% attack success on indirect prompt injection at 99% benign allow, and 0.205 ms median latency (under 0.1% overhead). Code and benchmarks released.

[4] **Right Knowledge, Wrong Answer: Test-Time Steering for Temporal Fact Conflicts in Open-Weight Language Models**
E. Hossain, *AAAI 2027, under review* arXiv:2606.20959

Defines Parametric Temporal Conflict, where a model holds the newer fact but prefers the outdated one, and releases a verified benchmark of 8,746 Wikidata transitions. Activation patching flips 72–85% of cases and localizes the preference to upper layers, and residual-stream steering beats random directions across four open-weight models.

[5] **Agreement Overstates Evidence: Error Dependence in LLM Judge Consensus**
E. Hossain, *ICLR 2027, under review* arXiv:2609.22512

Shows that LLM judges make correlated errors (mean pairwise correlation 0.21), so ten judges carry about as much information as 3.5 independent ones. Ignoring shared errors flips significance conclusions in up to 28% of comparisons; proposes estimating judge accuracy and shared mistakes from a small trusted set and choosing the voting rule on it.

[6] **ChannelGuard: Safe Models Do Not Compose into Safe Multi-Agent Systems**
E. Hossain, *ACL ARR, under review* arXiv:2607.19430

Shows on 2,100 traces (8 attack families, 3 backends) that apparent pipeline safety can come mostly from provider-side filters, and proposes training-free information-bottleneck gates on every inter-agent channel. Blocks tool poisoning 30/30 identically across backends and halves prompt-injection success (0.333 to 0.167) with GSM8K accuracy unchanged.

Research in Progress

- **Adaptive Compute Allocation for Vision-Language Models** *In preparation for CVPR 2027*
  Test-time compute in Qwen2.5-VL-3B: visual resolution, reasoning budgets, and adaptive routing, with cross-fitted oracle analysis of when dynamic allocation helps.
- **LLM Agents for Automated Model Deployment Optimization** *In preparation for ICML 2027*
  An LLM optimization agent for diffusion-model deployment, benchmarked against TPE, hierarchical Bayesian optimization, and random search, including hybrid LLM–Bayesian methods and held-out generalization.

Selected Technical Projects

- **MiniHealthLM** [GitHub]: Built a ∼600M-parameter Qwen-style Transformer from scratch (GQA, RoPE, RMSNorm) with medical-text preprocessing, PII filtering, custom BPE tokenization, and checkpointing. *PyTorch*
- **MLDriftGuard** [GitHub]: Installable Python package for numerical and categorical drift detection (KS and chi-squared tests) with visualizations and mitigation suggestions. *SciPy, pandas*

Technical Skills

Research	Reinforcement learning, offline RL, policy adaptation, LLM post-training, preference learning, uncertainty calibration, model evaluation, agent safety
ML / AI	Python, PyTorch, Hugging Face Transformers, RLHF, DPO, SFT, LoRA/QLoRA, RAG, FAISS, scikit-learn
Data / Tools	SQL, R, pandas, Git, Docker, Azure, GCP, multi-GPU training
Service	Reviewer, NeurIPS 2026 Workshop on Trustworthy AI for Good; ICML 2026 Workshop on Generative and Agentic AI for Biology